

Sample Surveys

Class 3

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Previously on Sample Surveys...

Recap

- **Sampling design:** A probability distribution over the collection of all possible samples.
- **Sampling scheme**
- **Simple random sampling (SRS):** Sample units one by one. Each unit has equal marginal probability of selection at any stage.
 - Simple random sampling with replacement (SRSWR)
 - Simple random sampling without replacement (SRSWOR)
- How do draw an SRS?
 - Classical techniques: Lottery method, Random number tables.
 - Modern techniques: Software packages (e.g., `sample` function in R).
- **Selection probability, Inclusion probability**
- Estimation of population mean $\bar{Y}$ under SRSWR
 - Estimator: Sample mean $\bar{y}$.
 - Sampling distribution of $\bar{y}$: Distribution induced by sampling.
 - $\bar{y}$ is **unbiased** for $\bar{Y}$.
 - **Sampling variance** of $\bar{y}$, $\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$ (**does not** depend on N other than through σ^2).
 - **Estimation** of sampling variance.

Simple Random Sampling *Without* Replacement (SRSWOR)

- Typically, an SRSWOR sample, one **gets rid of the order** of selected units.
 - E.g., $\{2, 3, 4, 7, 8\}$.
- What is the **sampling design** induced by SRSWOR?

SRSWOR: Sampling design

- Set $\mathcal{S}$: set of all unordered samples of size n from U , not allowing replacements.
- For an unordered sample $s \in \mathcal{S}$,

$$p(s) = \frac{1}{\binom{N}{n}}.$$

- The sampling design is **uniform** over $\mathcal{S}$.

- Selection probability: $\frac{1}{N}$ for all units, at every stage.
- Inclusion probability:
 - **First-order**: $\pi_j = P(\text{Unit } j \text{ is included in the sample}) = \frac{n}{N}$ for all $j \in U$.
 - **Second-order**: $\pi_{jj'} = P(\text{Units } j \text{ and } j' \text{ are included in the sample}) = \frac{n(n-1)}{N(N-1)}.$

Estimation under SRSWOR

- Parameter of interest: $\bar{Y}$.
- Turns out, $\mathbb{E}(\bar{y}) = \bar{Y}$ (why?).
- Sample mean is unbiased for the population mean under SRSWOR.
- What is its sampling variance?
- $\text{Var}(\bar{y}) = \frac{N-n}{N-1} \frac{\sigma^2}{n}$.
- Denote $S^2 = \frac{1}{N-1} \sum_{j=1}^N (Y_j - \bar{Y})^2$.

$$\text{Var}(\bar{y}) = \frac{N-n}{N} \frac{S^2}{n} = (1-f) \frac{S^2}{n}$$

- $f = \frac{n}{N}$ is called the sampling fraction.
- How to estimate this variance?
- Show that under SRSWOR, $\mathbb{E}(s^2) = S^2$.

$$\widehat{\text{Var}}(\bar{y}) = (1-f) \frac{s^2}{n}.$$

- The factor $(1-f)$ is called finite population correction.

More on SRSWR and SRSWOR

- Fix n . Which design is more efficient?
- For $n \geq 2$, $\text{Var}(\bar{y})$ under SRSWOR $<$ $\text{Var}(\bar{y})$ under SRSWR.
- Is SRSWR ever used?
 - In actual surveys, pretty rarely.
 - For large N , the difference between SRSWOR and SRSWR is negligible.
 - SRSWR can save computational time and/or computer memory.
 - An important application of SRSWR: **Bootstrap** methods.
- Another alternative estimator under SRSWR: Look at all the **distinct** units.
 - Effective sample size: n_{eff} : Number of distinct units in the sample.
 - n_{eff} is a **random variable**.
 - What is $\mathbb{E}(n_{\text{eff}})$?
 - $\bar{y}_{\text{eff}}$: Sample mean of Y across the distinct units in the sample.
 - $\mathbb{E}(\bar{y}_{\text{eff}}) = \bar{Y}$. (why?)
 - SRSWOR with n units vs SRSWR with distinct units.

Other population parameters

- Estimation of population total of Y : $\sum_{j=1}^N Y_j$.
 - **Example:** Estimating the total wheat production across all farms in a region to determine if food imports are needed.
- Estimation of population variance of Y : $\sigma^2 = \frac{1}{N} \sum_{j=1}^N (Y_j - \bar{Y})^2$.
 - **Example:** Estimating the variance of household wealth within a district to quantify economic inequality.
- Consider two study variables X and Y . Estimation of population covariance of X and Y : $\frac{1}{N} \sum_{j=1}^N (X_j - \bar{X})(Y_j - \bar{Y})$.
 - **Example:** Estimating the covariance between out-of-pocket healthcare expenditure (X) and household debt (Y) to design effective insurance policies.

Estimating population proportions under SRS

- What fraction of young adults in a state are currently unemployed?
 - What proportion of street vendors now accept digital payments (like UPI) instead of just cash?
 - What fraction of small business owners have to rely on local moneylenders instead of formal bank loans?
 - What percentage of rural clinics completely run out of basic antibiotics at least once every three months?
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- How to formalize these questions?
 - Binary study variable $Y \in \{0, 1\}$, denoting some characteristic (e.g., being unemployed).
 - $Y_j = \mathbb{1}(\text{unit } j \text{ in the population possesses the characteristic})$
 - In the first example, $Y_j = 1$ if unit j is unemployed, $Y_j = 0$, otherwise.
 - Parameter of interest: population proportion $P = \frac{1}{N} \sum_{j=1}^N Y_j = \bar{Y}$.
 - Back to the previous case of population mean estimation.
 - Estimation? Inference?

Other measures of accuracy

- Measures of **sampling variability** of an estimator, i.e., variability with respect to the sampling distribution of the estimator.
- Standard error of an estimator, say $\bar{y}$: $SE(\bar{y}) = \sqrt{\text{Var}(\bar{y})}$.
 - The estimated standard errors are commonly reported in surveys.
 - E.g., under SRSWOR, $\widehat{SE}(\bar{y}) = \sqrt{\widehat{\text{Var}}(\bar{y})} = \sqrt{1-f} \times \frac{s}{\sqrt{n}}$.
- Coefficient of variation (CV) or Relative standard deviation (RSD)
 - Measures the extent of variability **in relation to the truth**.
 - CV of an estimator (also called **relative standard deviation or RSD**): Ratio of the SE of the estimator to the parameter of interest.
 - $CV(\bar{y}) = \frac{SE(\bar{y})}{\bar{y}}$.
 - Estimated $CV(\bar{y})$: $\widehat{CV}(\bar{y}) = \frac{\widehat{SE}(\bar{y})}{\bar{y}}$.

Statistics Canada guidelines

- $CV \leq 16.6\% \implies$ reliable. ✓
- $16.6\% < CV \leq 33.3\% \implies$ give warning. ▲
- $CV > 33.3\% \implies$ unreliable. ✗